WORKED SOLUTION

STEP II 2020, Question 8

Pure • Polynomials and Integration

STEP II 2020, Question 8

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Because $\mathrm{F}' = \mathrm{f}$, every root of f is a turning point of F, and each area condition becomes a statement about the value of F at one of those turning points. Once we know F has double roots at 0 and c, the sum $\mathrm{F}(x) + \mathrm{F}(c - x)$ collapses to a single sign-carrying factor, $c - 2h$. Comparing that sign with what we know about the largest and smallest values of F on $[0, c]$ forces the symmetry.

Solution

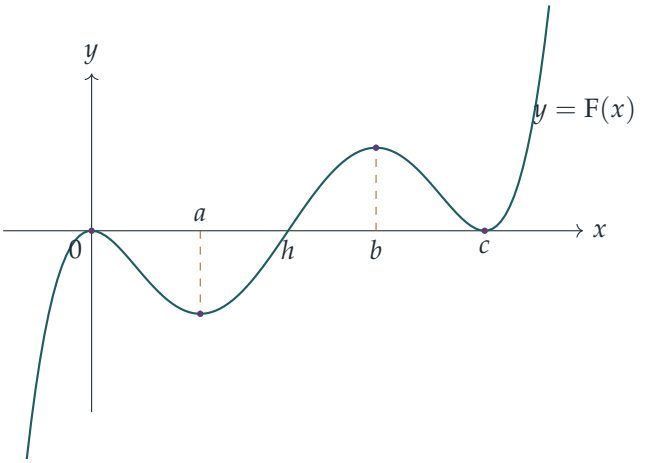
Since f is monic with roots $0, a, b, c$, we have $\mathrm{f}(x) = x(x-a)(x-b)(x-c)$. So $\mathrm{f} < 0$ on $(0, a)$, $\mathrm{f} > 0$ on (a, b), $\mathrm{f} < 0$ on (b, c), and $\mathrm{f} > 0$ for $x < 0$ and $x > c$. Write A for the area between 0 and a. The conditions say that the area between b and c is A and the area between a and b is $2A$, with $A > 0$.

Part (i)

Because $\mathrm{F}(x) = \int_0^x \mathrm{f}(t)\,\mathrm{d}t$, the integral over an interval where $\mathrm{f} < 0$ is minus the area there. So

$$\mathrm{F}(0) = 0, \qquad \mathrm{F}(a) = -A, \qquad \mathrm{F}(b) = -A + 2A = A, \qquad \mathrm{F}(c) = A - A = 0.$$

Since $\mathrm{F}' = \mathrm{f}$, the turning points of F are at $x = 0, a, b, c$. The sign of f changes from $+$ to $-$ at 0 and at b, giving maxima there, and from $-$ to $+$ at a and at c, giving minima. F is a quintic with positive leading coefficient $\frac{1}{5}$, so $\mathrm{F}(x) \to \pm\infty$ as $x \to \pm\infty$.



The curve has a maximum at $(0,0)$, a minimum at $x=a$ with $y=-A$, a maximum at $x=b$ with $y=A$, and a minimum touching the axis at $(c,0)$.

The form of F. F is a polynomial of degree 5 with leading coefficient $\frac{1}{5}$. At $x=0$ we have $\mathrm{F}(0)=0$ and $\mathrm{F}'(0)=\mathrm{f}(0)=0$, so $x=0$ is a root of multiplicity at least 2. In the same way $\mathrm{F}(c)=0=\mathrm{F}'(c)$, so c is a root of multiplicity at least 2. That accounts for four of the five linear factors:

$$\mathrm{F}(x)=\tfrac{1}{5}x^2(x-c)^2(x-h)$$

for some real h. Also, $\mathrm{F}(a)=-A<0<A=\mathrm{F}(b)$, so F has a root strictly between a and b. Neither 0 nor c lies in (a,b), so this root must be h. Hence $a<h<b$, and in particular $0<h<c$.

The sum. Note that $(c-x)^2((c-x)-c)^2=x^2(x-c)^2$. So

$$\begin{aligned}\mathrm{F}(x)+\mathrm{F}(c-x)&=\tfrac{1}{5}x^2(x-c)^2[(x-h)+(c-x-h)]\\&=\tfrac{1}{5}x^2(x-c)^2(c-2h).\end{aligned}$$

Part (ii)

On $[0,c]$, F is strictly decreasing on $[0,a]$, strictly increasing on $[a,b]$ and strictly decreasing on $[b,c]$. So on $[0,c]$ the smallest value is $\mathrm{F}(a)=-A$: the only other candidate is the endpoint value $\mathrm{F}(c)=0>-A$. This minimum is attained only at $x=a$. So for $0\le x\le c$,

$$\mathrm{F}(b)+\mathrm{F}(x)\ \ge\ \mathrm{F}(b)+\mathrm{F}(a)=A-A=0,$$

with strict inequality when $x\neq a$.

Ask yourself Which value of x makes $\mathrm{F}(b)+\mathrm{F}(x)$ look like the sum from part (i)?

Take $x = c - b$, which lies in $[0, c]$ because $0 < b < c$. By part (i),

$$\mathrm{F}(b) + \mathrm{F}(c-b) = \tfrac{1}{5}b^2(b-c)^2(c-2h).$$

If $c - b \neq a$, the left-hand side is strictly positive. Since $\frac{1}{5}b^2(b-c)^2 > 0$, it follows that $c - 2h > 0$. So either $c - b = a$ or $c > 2h$, as required.

Now run the same argument the other way up. On $[0, c]$ the largest value of F is $\mathrm{F}(b) = A$, beating the endpoint values $\mathrm{F}(0) = \mathrm{F}(c) = 0$, and it is attained only at $x = b$. So for $0 \leq x \leq c$,

$$\mathrm{F}(a) + \mathrm{F}(x) \leq \mathrm{F}(a) + \mathrm{F}(b) = 0,$$

with strict inequality when $x \neq b$. Taking $x = c - a \in [0, c]$ gives $\frac{1}{5}a^2(c-a)^2(c-2h) \leq 0$, strictly if $c - a \neq b$. So either $c - a = b$ or $c < 2h$.

If $c \neq a + b$, both alternatives give $c > 2h$ and $c < 2h$, which is impossible. Hence $c = a + b$. Then $c - b = a$, so

$$0 = \mathrm{F}(b) + \mathrm{F}(a) = \mathrm{F}(b) + \mathrm{F}(c-b) = \tfrac{1}{5}b^2a^2(c-2h).$$

Because $a, b \neq 0$, this gives $c = 2h$.

Part (iii)

With $h = \frac{1}{2}c$, we have $\mathrm{F}(x) = \frac{1}{5}u\left(x - \frac{1}{2}c\right)$, where $u = x^2(x-c)^2$, so that $u' = 2x(x-c)(2x-c)$. Differentiating with the product rule,

$$\begin{aligned}
\mathrm{f}(x) = \mathrm{F}'(x) &= \tfrac{1}{5}\left[2x(x-c)(2x-c)(x-\tfrac{1}{2}c) + x^2(x-c)^2\right] \\
&= \tfrac{1}{5}x(x-c)[(2x-c)^2 + x(x-c)] \\
&= \tfrac{1}{5}x(x-c)(5x^2 - 5cx + c^2) \\
&= x(x-c)(x^2 - cx + \tfrac{1}{5}c^2).
\end{aligned}$$

(So $a, b = \frac{1}{2}c \mp \frac{c}{2\sqrt{5}}$, which do add up to c.)

> **Ask yourself** The roots are symmetric about $x = \frac{1}{2}c$. Can we use that to make the second derivative easy?

Yes. Put $y = x - \frac{1}{2}c$. Then $x(x-c) = y^2 - \frac{1}{4}c^2$ and $x^2 - cx + \frac{1}{5}c^2 = y^2 - \frac{1}{20}c^2$, so

$$\mathrm{f} = (y^2 - \tfrac{1}{4}c^2)(y^2 - \tfrac{1}{20}c^2) = y^4 - \tfrac{3}{10}c^2y^2 + \tfrac{1}{80}c^4.$$

Since $\mathrm{d}y/\mathrm{d}x = 1$,

$$\mathrm{f}''(x) = 12y^2 - \tfrac{3}{5}c^2,$$

which is zero exactly when $y^2 = \frac{1}{20}c^2$. At those points the factor $y^2 - \frac{1}{20}c^2$ of f vanishes, so $\mathrm{f} = 0$. Each is a simple root of the quadratic f'' in y, so f'' changes sign there, and they are genuine points of inflection. So both points of inflection lie on the x-axis. In fact they are the points $(a, 0)$ and $(b, 0)$.

Checks

- Take $c = 2$, so that $a = 1 - \frac{1}{\sqrt{5}} \approx 0.5528$ and $b \approx 1.4472$. Numerical integration of f gives areas 0.05724, 0.11449 and 0.05724 over $[0, a]$, $[a, b]$ and $[b, c]$: the outer two are equal, and each is half the middle one.
- Expanding $x(x-a)(x-b)(x-c)$ with $a + b = c$ and $ab = \frac{1}{5}c^2$ reproduces the expression in part (iii).
- With $c = 2$, a finite-difference estimate of f'' at $x = a \approx 0.5528$ is zero to within rounding, and it changes sign between $x = 0.54$ and $x = 0.56$.

Common slips

- Forgetting that $\mathrm{F}'(0) = \mathrm{F}'(c) = 0$ as well as $\mathrm{F}(0) = \mathrm{F}(c) = 0$, and so missing the double roots.
- Not justifying $0 < h < c$: the sign change of F between a and b is what places h.
- Using only the first inequality in part (ii). It gives "$c = a + b$ or $c > 2h$", which alone is not enough. The reversed inequality with $\mathrm{F}(a)$ is needed to close the argument.

Beyond STEP

The argument in part (ii) uses the extreme value theorem: a continuous function on a closed interval attains a greatest and a least value, and here we also know exactly where. The quartic's symmetry about $x = \frac{1}{2}c$, which the area conditions force, is an example of how integral conditions can pin down the shape of a polynomial. Abbott's *Understanding Analysis* gives a careful treatment of the extreme value theorem and of the fundamental theorem of calculus, which links F and f.

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